CHAPTER 5

Liquid Crystals

LC
↓
S: → cigar, layers
N: - 1-dimen liq.
C: - plate shaped

5.1 Introduction

Liquid crystals are the intermediate phases between liquid and crystal. Liquid crystal displays [LCDs] do not generate light energy, but simply alter or control the existing light to make selected areas appear bright or dark. Liquid crystals have orientational order but lack positional order. The material that used in liquid crystal display is 4-methoxy - 4′n-butyl benzylidene aniline (MBBA) molecules. It can act as a liquid crystal between the temperatures of 21°C to 48°C. It has an elongated rod-like structure. There are three phases in liquid crystals. They are smectic, nematic and cholestic. In smectic phase the molecules are cigar shapped and are arranged in layers. The molecules can move forward and backward but not up and down. In nematic phase all the molecular axes are parallel to each other, but it is not a layered structure. This can be considered as a one-dimensional liquid. In the cholestic phase the molecules are plate-shapped and this is a stack of thin layers. As one goes down the stack, the direction of orientation rotates in the form of a screw. This phase of crystals possess double refraction. The other liquid crystal is pentyl cyano biphenyl. This has nematic form in the temperature range of 18°C to 35°C.

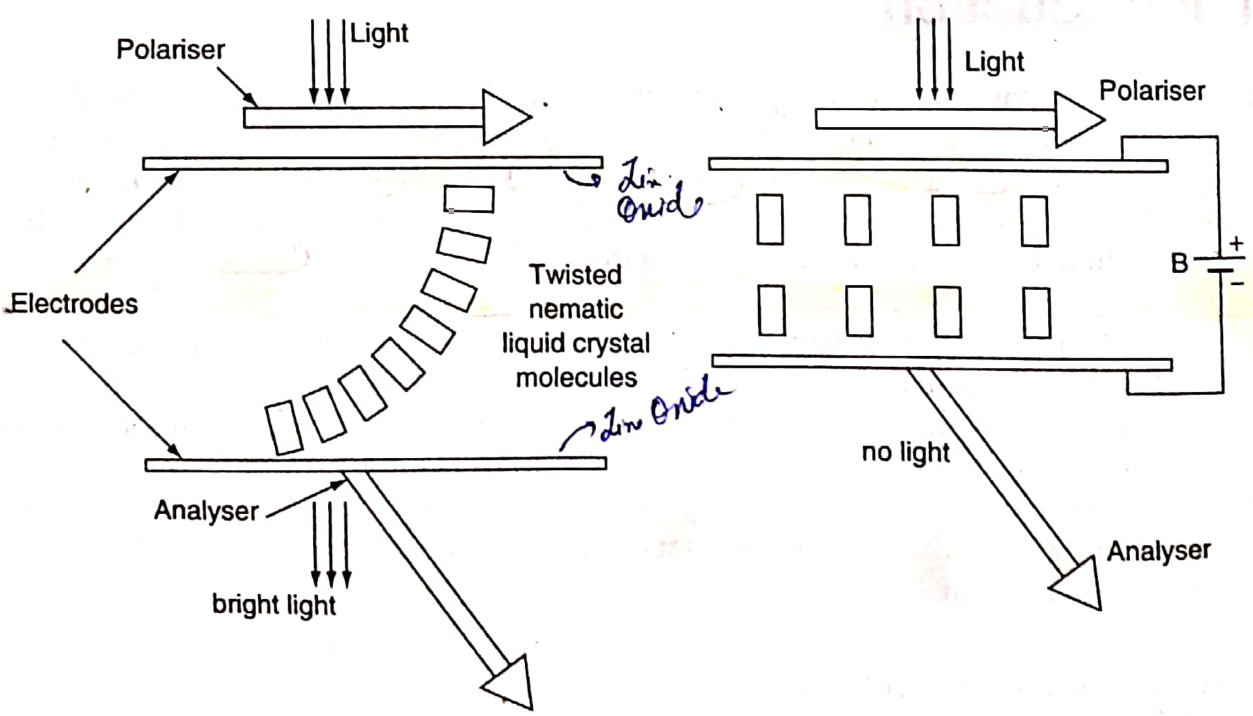
Properties of liquid crystals:

1. Liquid crystals are very sensitive to temperature, electric field, mechanical stress, etc. Change in any of the above parameters causes change in the optical properties of liquid crystals.
2. As temperature changes the cholestric phase can change the colour of radiation. This fact is used to measure the temperature of children, and also in thermography to detect breast cancer.
3. Nematic phase is widely used for displays. The display is based on the change of polarisation during the application of electric field.

5.2 Liquid crystal display system

There are two types of liquid crystal displays (i) Dynamic scattering display and (ii) Twisted nematic field effect display. The dynamic scattering display is not presently used because of its short life time and larger power consumption during operation. In case of twisted nematic field effect disply, a thin layer of liquid crystal material of 10–20 μm thick is kept in between two glass plates coated with transparent tin oxide on the inner side surfaces, which acts as electrodes as shown in Fig. 5.1(a). In the absence of applied electric field, the top glass plate is rotated through 90°, this causes the liquid crystal molecules also to be twisted through 90°. Above the top glass plate, a polariser and below the bottom glass plate, an analyser are kept in crossed positions. When light is allowed to pass through the liquid crystal cell and through the crossed polariser and analyser, the cell appears bright due to additional phase difference introduced by twisting. When an electric field is applied, the liquid crystal molecules orient themselves parallel to the field direction as shown in Fig. 5.1(b). Hence the cell appears dark due to the crossed polariser and analyser.

Figure 5.1 (a) Twisted nematic field effect display in the absence of applied electric field (b) Twisted nematic field effect display under applied electric field



5.3 Merits and demerits of LCDs

The following are the merits and demerits of LCDs:

(i) The threshold voltage of twisted nematic LCDs is low [few volts].
(ii) Higher contrast.

(iii) The use of two polarizers reduces the brightness of the display considerably.
(iv) The twisted nematic LCDs work best when viewed straight, whereas the degradation of performance as the viewing angle is changed.
(v) The switching times for nematic LCDs are not fast [0.02 to 0.05 seconds].
(vi) The nematic LCDs would twist in opposite directions when no voltage was applied, so this gives patchy appearance. To oversome this a chiral nematic liquid crystals are used.

5.4 Applications of liquid crystals

(i) Liquid crystals have been used in displays.
(ii) The colour of liquid crystal vary with temperature, hence the liquid crystals have been used in the construction of sensitive thermometers to measure temperature of children.

5.5 Metallic glasses-Types of metallic glasses

Glass is a non-crystalline solid obtained by freezing a liquid, so that it is in vitreous state. The chemical formula of an oxide glass is A_mB_nO. In this formula m and n are usually not integers and represent simply the number of atoms A and B per oxygen atom. Metallic glasses are obtained by ultra fast quenching of liquid metal alloys at cooling rate at 10^6 to 10^8K per second. The metallic glasses are divided into two types. They are (a) Metal-Metalloid alloys and (b) Metal-Metal alloys.

(a) **Metal-Metalloid alloys:** It has the general formulae $[M_1, M_2, \ldots]_{80}\,[m_1, m_2, \ldots]_{20}$. where M_i is a transition metal like Au, Pd, Pt, Fe, Ni or Mn and m_i is metalloid like Si, Ge, P, C or B. The 80/20 ratio is approximate.

(b) **Metal-Metal alloys:**. In these the ratio of atoms is approximately 50% one of the metals is generally a transition metal. For example: $Mg_{65}Cu_{35}$, $Au_{55}Cu_{45}$, $Sn_{90}Cu_{10}$, $Zr_{72}Co_{28}$, $Zr_{50}Cu_{50}$, $Ni_{60}Nb_{40}$.

5.6 Properties of metallic glasses

The various properties of metallic glasses are given below.

(i) **Structural properties:** The properties which are sensitive to structure and to impurities are transport properties. They are electrical conductivity and thermal conductivity. Also the various losses i.e. dielectric and viscoelastic losses depend structure of metallic glass molecules. The structure independent properties are the density, elastic constants, specific heat, dielectric permittivity, etc.

(ii) **Mechanical properties:** Metallic glasses posses high mechanical strength and hardness, radiation resistance, plastic deformation occurs at high stress levels before fracture.

(iii) **Electrical properties:** At room temperature, the electrical resistivity of the metallic glasses is higher than the crystalline metals or alloys. Due to the high resistivity the eddy current loss is very small. Glassy metals are extensively used in electronic devices. They are used in audio and video digital recording. The metallic glasses are also used in recording catridges, amorphous heads for audio and computer tape recording.

(iv) Magnetic properties: Some of the metallic glasses possess soft magnetic properties and some other possess hard magnetic properties, hence they find large number of applications in industries. Ferrous glasses are easily magnetised upto saturation limit. The core losses of soft metallic glasses are very less nearly 1/6 to 1/4 of the convensional materials. Due to the extremely low magnetic losses and zero magnetostruction these materials provides substitutes for core transformer steels. These materials have very low coercivities and posses square loop characteristies shows energy saving.

(v) Chemical properties: The metallic glasses possess high chemical corrosion resistance, for example the formation of protective oxide film in chromium containing glasses resists corrosion. Chemically metallic glasses are more resistant than their crystalline counter parts. The metallic glasses can act as catalysts beçause amorphous state is more active than the crystalline state from the catalytic point of view.

5.7 Shape memory alloys

A class of metal alloys which have the ability to return back from deformed shape to its previous shape [the alloy appears to have memory] by applying heat or by releasing stress in some cases are called shape memory alloys (SMA). These materials remember their shape even after severe deformation. SMAs were discovered in 1932 and many of their applications were not known until 1961. The best example for SMAs is nickel-titanium alloy, other examples are Cu-Al-Ni, Cu-Zn-Al, Co-Ni-Al, Mn-Cu, Ti-Pd-Ni, Ni-Ti-Hf, etc. The SMAs show shape memory effect and pseudoelasticity or superelasticity. These are described below.

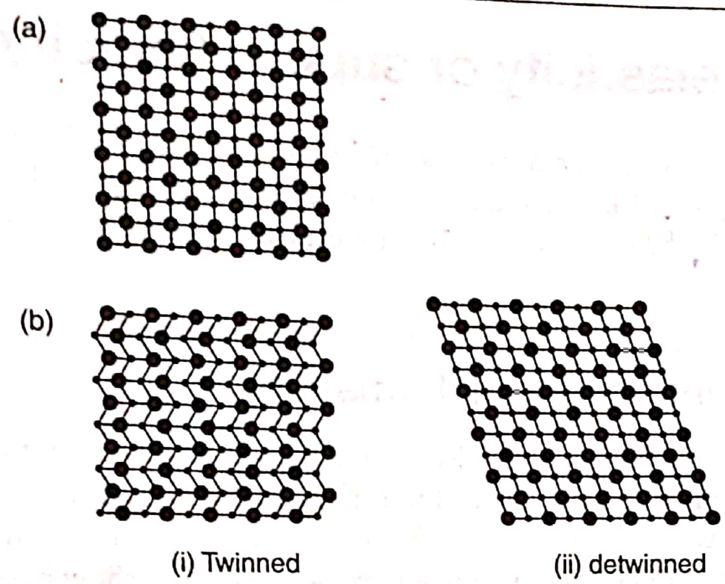
5.8 Shape memory effect

In this phenomenon a shape memory alloy is deformed at lower temperature and reverts to its undeformed original shape when heated to higher temperature. The materials which exhibit shape memory effect on heating only are called one way shape memory alloys. On the other hand the materials which show a shape memory effect on heating and on cooling are called two way shape memory alloys. The shape memory effect occurs in some metallic alloys due to the change in their crystalline structure. The change in crystal structure is due to the change in temperature and stress of the material. The SMA shows two solid phases due to their different crystal structures. The shape memory effect has been explained by considering a bar of Ni-Ti alloy material in austenite phase (hard). The high temperature phase of Ni-Ti is called austenite [microscopically they possess small platelet structure].

This has a highly symmetric cubic crystal structure as shown in Fig.5.2(a). The bar is allowed to cool well below the phase transition temperature. The low temperature phase of Ni-Ti alloy is called martensite [microscopically they possess needle like structure]. This has a low symmetry monoclinic crystal structure. The martensite can be in one of two forms, twinned and detwinned as shown in

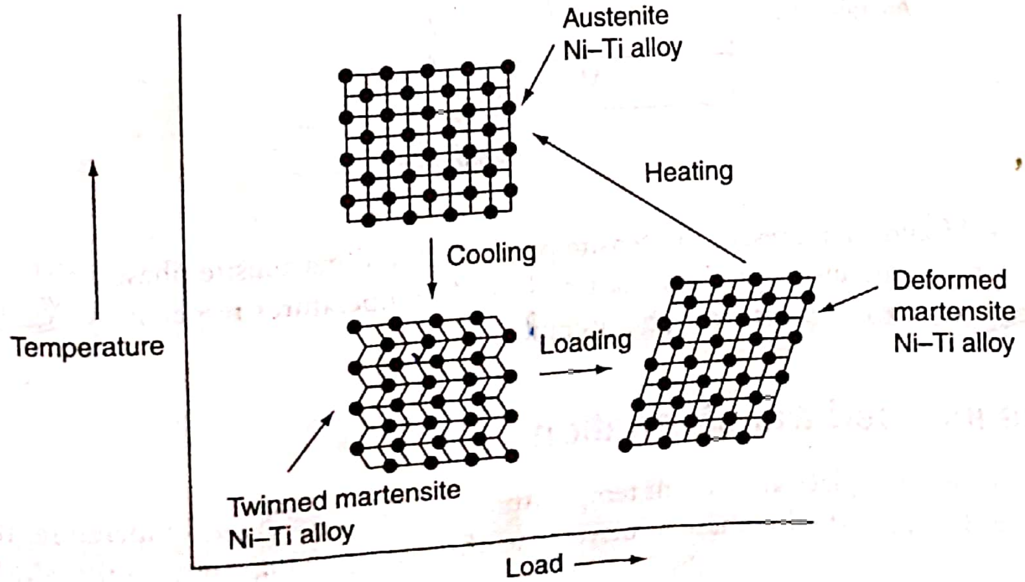
Fig. 5.2(b). On applying external stress the material go from twinned to detwinned form and the material stretches beyond its normal elastic limit. This is called pseudo elasticity.

Figure 5.2 (a) Austenite phase of Ni–Ti alloy; (b) Martensite phase of Ni–Ti alloy



At low temperature the bar is plastically deformed [say by bending the bar]. Again by heating the bar to above its phase transition temperature the bar will come back to its original configuration. The twinned martensite structure show shorter length whereas the detwinned or deformed martensite structure is longer. On heating, this material changes from martensite to austenite phase with increase in length. The Ni-Ti alloy possess martensite transition temperatures between 62°C to 72°C and the austenite transition temperatures between 88°C to 98°C. Schematically, the crystallographic formation of martensite and reversion to austenite or on heating is shown in Fig. 5.3.

Figure 5.3 Shape memory effect in Ni–Ti alloy



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Some SMAs, in martensite phase exhibit body centred tetragonal crystal structure. These show longer length along Z-direction of the crystal in martensite phase than in austenite phase. On heating this material changes into austenite phase with shrinkage in length. These materials will not show twinning and pseudoelasticity.

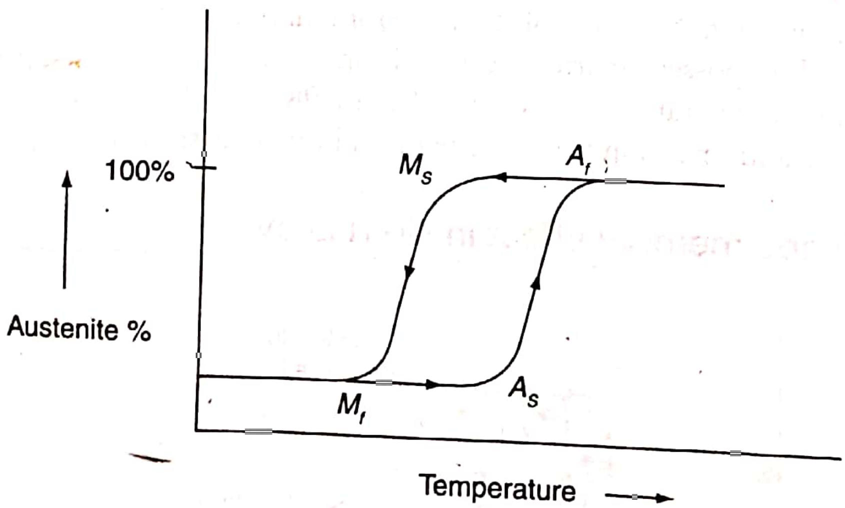
5.9 Pseudo elasticity or super-elasticity

This refers to the ability of a shape memory alloy (SMA) to return to its original shape after substantial deformation upon unloading. The phase transitions in SMAs depends on the temperature variations or on stress variation or both. The first two cases are explained in this topic. We come across pseudo elasticity in second case.

(a) Temperature induced transformation

The phase transformation in SMAs takes place not at a particular temperature but over a range of temperatures as shown in Fig. 5.4. The range of transformation temperature from martensite phase to austenite phase during heating is higher than that for the reverse transformation upon cooling. The transformations on heating and on cooling do not overlap and hence show hysteresis.

Figure 5.4 Temperature induced phase transition in SMA

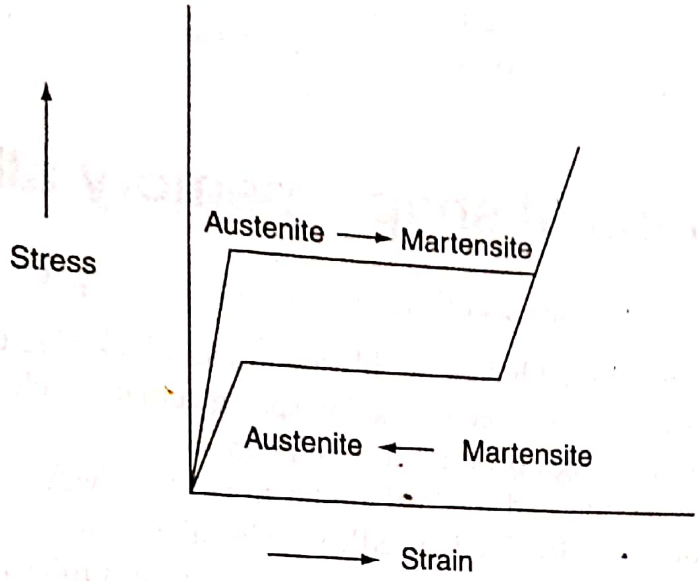


In Fig. 5.4, M_s and M_f represent martensite phase start and martensite phase finish temperature. Also A_s and A_f represent austenite phase start and finish temperatures respectively. The hysteresis is found to depend on the composition of the system.

(b) Stress induced transformation

This transformation takes place at constant temperature, usually above the A_f temperature. The material can be transferred from austenite phase to martensite phase by applying stress on the SMA as shown in Fig. 5.5.

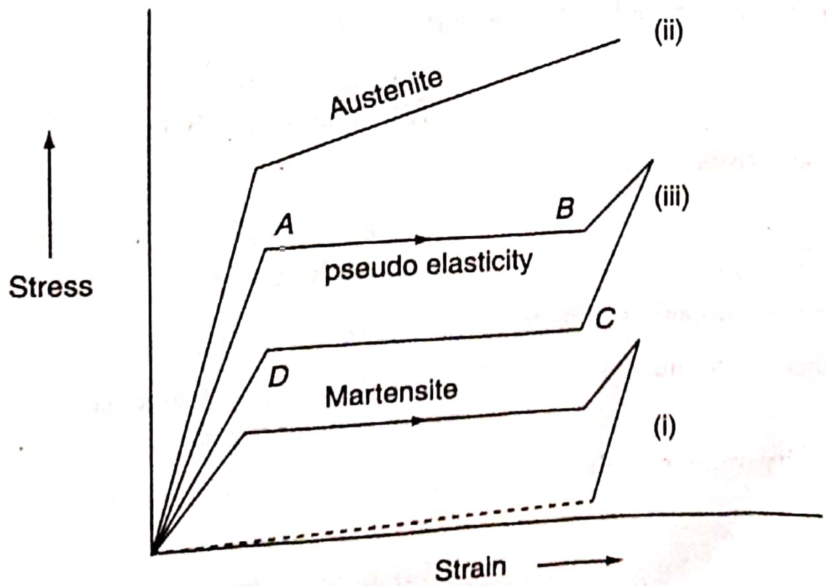
Figure 5.5 Stress induced transformation in SMA



The pseudoelasticity is based on stress induced martensitic transformation to austenitic above the temperature A_f, so that the material comes back to its original shape. Here the material exhibits extremely elastic is known as pseudoelasticity or superelasticity. This pseudoelasticity is non-linear and depends on temperature and strain.

Fig. 5.6 shows the stress strain behaviour of Ni-Ti alloy at below, within and above its transformation temperature range. At below transformation temperature range the curve (i) is obtained, it is the martensite phase. This easily deformed to several percent strain even at low stress. In this curve the dotted portion represents the rememberence of original shape of the material when stress is removed. The curve (ii) is drawn at above its transformation temperature, at this temperature the austenite phase is found to have higher yield and flow stresses. Curve (iii) is drawn within its transformation termperature, at this temperature the material shows shape memory effect with high elastic property known as pseudo

Figure 5.6 Stress–strain behaviour of SMA at different temperatures



elastic behaviour because the martensite phase is obtained with stress (curve AB) again on unloading the material is found to return to its original shape (curve CD) without applying heat. This is a pure mechanical behaviour and is refered to as pseudo elasticity or super plasticity. The pseudo elasticity behaviour is applied in making eye glass frames, medical tools and cellular phone antennae's.

5.10 Applications of shape memory alloys (SMA)

The following are some of the applications of the safe memory and superelastic alloys:

(i) The SMAs are used as fine tuned helicopter blades and as a lock ring electrical connectors.

(ii) In automobile industry SMAs are used in making spring actuators, clutch systems, thermostates, oil pressure control unit and high pressure sealing plugs.

(iii) In the field of bio-engineering and medical applications the SMAs are used as (a) dental arch wires and (b) blood clot filters. The Ni-Ti needle wire localizers are used to locate breast tumours. The SMAs are used in designing microsurgical instruments and microgrippers, etc.

(iv) The shape memory alloys are used in making eye glass frames, in robotic actuators, in toys and ornamental goods, couplers and fasteners, fixed safety valves, temperature sensitive SMA valves etc.